A FINITE MODULE WHOSE K-TH FITTING IDEAL IS PRINCIPAL IS GENERATED BY K ELEMENTS AFTER REMOVING ITS F-TORSION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot (Xy-pic setup plus theorem declarations and mathematical macros). Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Allowed standard prerequisites (editorial).

- Fitting ideal of a finite module and its description through the minors of a presentation matrix. defined in the same section, source lines 1320-1578, before the counted statement. Supplies the ideal whose being principal is the hypothesis.
- Nakayama's lemma in the form that a set of elements generating a principal ideal over a local ring contains a generator. the author cites Algebra, Lemma algebra-lemma-NAK at proof lines 1591-1593. Allows the Fitting ideal to be generated by a single chosen minor.
- Existence of a right inverse for a matrix with an invertible minor, up to the factor given by that minor. the author cites Algebra, Lemma algebra-lemma-matrix-right-inverse at proof lines 1601-1603. Produces the matrix whose columns give the combinations annihilated by f.

Source reference. Original lemma 080Z: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10199–10199. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 00DV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 512–512. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 080R: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10191–10191. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let R be a local ring. Let M be a finite R-module. Let $k \geq 0$. Assume that $Fit_k(M) = (f)$ for some $f \in R$. Let M' be the quotient of M by $\{x \in M \mid fx = 0\}$. Then M' can be generated by k elements.*

Proof. Choose generators $x_1, \ldots, x_n \in M$ corresponding to the surjection $R^{\oplus n} \to M$. Since R is local if a set of elements $E \subset (f)$ generates (f), then some $e \in E$ generates (f), see Algebra, Lemma 00DV. Hence we may pick $z_1, \ldots, z_{n-k}$ in the kernel of $R^{\oplus n} \to M$ such that some $(n-k) \times (n-k)$ minor of the $n \times (n-k)$ matrix $A = (z_{ij})$ generates (f). After renumbering the x_i we may assume the first minor $\det(z_{ij})_{1 \leq i,j \leq n-k}$ generates (f), i.e., $\det(z_{ij})_{1 \leq i,j \leq n-k} = uf$ for some unit $u \in R$. Every other minor is a multiple of f. By Algebra, Lemma 080R there exists a $n - k \times n - k$ matrix B such that

$$AB = f \begin{pmatrix} u1_{n-k \times n-k} \\ C \end{pmatrix}$$

for some matrix C with coefficients in R. This implies that for every $i \leq n-k$ the element $y_i = ux_i + \sum_j c_{ji} x_j$ is annihilated by f. Since $M / \sum Ry_i$ is generated by the images of $x_{n-k+1}, \ldots, x_n$ we win. $\square$